Name: Anthony Tom, Andrew ID: amtom, Section: S

1. **Project Title and Description**
    1. Project Title: Classic Chess
    2. Project Description: I plan to make a traditional chess game with a wide variety of features using cmu_graphics. There will be two modes, Single Player, where a player can play against an AI, and Multiplayer, where two players can play against each other. I plan to make the chess game visually appealing with smooth animations and include a timer for every player. For the AI, I plan to utilize recursion and backtracking so that the AI can choose a piece and make the first valid move, which would be Easy Mode. For Medium Mode, I plan to use a modified version of the minimax algorithm, which utilizes recursive trees to create the next optimal move for the AI. I plan to take the average minimax scores of possible starting moves and move the piece with the highest average score. At depths greater than 2, the program will run slowly, so the minimax bot may not make the best moves. For Hard Mode, I plan to utilize a bot that immediately captures random pieces. For Extreme Mode, I plan to utilize a bot that always captures the piece with the highest value. Additionally, I will include the special chess moves like en passant, castling, and promotions. Furthermore, I will use recursion to test if there is a checkmate or stalemate. This description has been updated.
2. **Similar Projects**
    1. Chess AI by Ares Kapoor https://www.youtube.com/watch?v=Ijc2SyD4754. Ares Kapoor made a chess game with two modes: Multiplayer and Play Against AI. In the video, he stated that he used the minimax algorithm for both the medium and hard modes of the AI. A flat 2D chess board was made to display the game. When a piece is clicked, green dots are displayed on the board that show possible places the piece can be placed, enabling players to understand legal moves. Recursion can be implemented to test the next possible moves for every piece.
    2. AI Chess by Christian Angelov https://www.youtube.com/watch?v=uJhGUxTs5sU. Christian Angelov made a chess game where a player can play against another player or against an AI. However, unlike Ares's project, Christian's project did not have difficulty modes for the AI. A notable feature of AI Chess is that key presses can be used to promote pawns to other pieces when they reach the end of the board.
    3. Yuvi Chess Trainer by Yuvanshu Agarwal https://www.youtube.com/watch?v=gSP4xnh7674. The program includes all traditional chess moves, including regular piece moves, promotions, en passant, and castling. The program analyzes the moves that the player makes. Yuvanshu utilized neural networks to analyze hundreds of thousands of grandmaster positions. The minimax algorithm was also used for the computer to select the optimal next move. I believe it is a good idea for the chess program to analyze chess positions from professional games.

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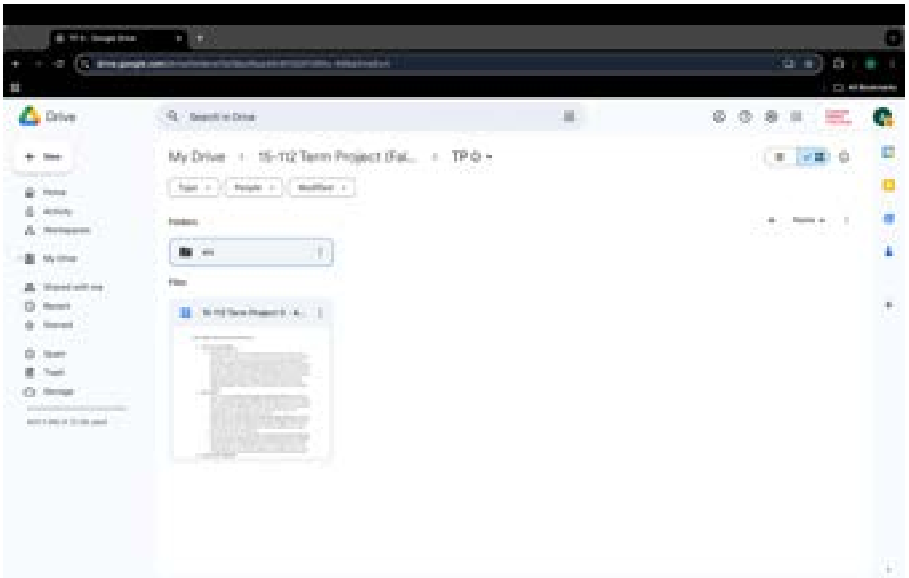
3. **Version Control / Backup Plan**
    1. I plan to store my files in Google Drive, since I can easily access my programs using my Google accounts. I will download the Python files consistently and add these files to src folders in Google Drive. I will also add dates in programs and in folder names. If I become sufficiently familiarized with Github, I would like to store my files in a private git repository.



4. **Tech List**
    1. The only modules I plan to use in my program are built-in Python modules. I will use the built-in math module for any calculations that I need to make. I will also use the built-in random module to generate random moves. Furthermore, I plan to utilize the built-in time module to create a chess timer for both players.
    2. I also plan to use the python-chess module to understand the features that I can implement in my game. However, I will likely not use this module directly in my program because I plan to create all of the features of the chess game without external modules.